CO2-AT1-Analytical Problem Solving

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Slot :- A

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1)A) Thresholding and intensity-Level Splicing - PCB Inspection :-

Given:-

Original pixel values:

$$r = [25, 60, 95, 130, 165, 200, 235]$$

Intensity band = 90-170

(a) Apply binary threshold transformation :-

$$S = \begin{cases} 255 & r \geq 128 \\ 0 & r < 128 \end{cases}$$

Input(r)	output(s)
25	0
60	0
95	0
130	255
165	255
200	255
235	255

(b) Apply intensity-level slicing:-

Rule :-

- If the intensity is between 90 and 170, output = 255.

• otherwise, retain the original intensity.

Input (r)	output (s)
25	25
60	60
95	255
130	255
165	255
200	200
235	235

(c) calculate number and Percentage of highlighted Pixels.

Number of highlighted Pixels = 3

Total pixels = 7

$$\text{Percentage} = 3/7 \times 100 = \boxed{42.86\%}$$

(d) Calculate mean intensity

original mean:

$$\text{Mean} = \frac{\Sigma r}{N}$$

$$= \frac{25+60+95+130+165+200+235}{7}$$

$$= 910/7 = \boxed{130}$$

Sliced output mean:-

$$= \frac{25+60+255+255+255+200+235}{7}$$

$= \frac{1285}{7} = \boxed{183.57}$

(e) which transformation is more appropriate?

Intensity-level slicing is more appropriate for isolating solder joints because it highlights only the required intensity range of 90-170.

2,A) Piecewise-Linear contrast stretching X-Ray Image-Enhancement :-

Given:

$(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$

$L = 256$

Input pixels:-

$r = [30, 70, 90, 110, 140, 200]$

Assume the standard endpoints (0,0) and (255,255)

(a) Derive the equations and slopes.

The slope formula is:

$$m = \frac{s_2 - s_1}{r_2 - r_1}$$

Segment 1: 0-70

$$m_1 = \frac{20 - 0}{70 - 0} = \frac{20}{70} = 0.286$$

Equation:-

$$S = 0.286r$$

Segment 2: 70 - 140

$$m_2 = \frac{230 - 20}{140 - 70}$$

$$= \frac{210}{70} = 3$$

Equation: $S - 20 = 3(r - 70)$

$$S = 3r - 190$$

Segment 3:- 140 - 255

$$m_3 = \frac{255 - 230}{255 - 140}$$

$$= \frac{25}{115} = 0.2174$$

Equation:- $S = 230 + 0.2174(r - 140)$

(b) Calculate transformed values.

For input pixel $r = 30$

$$S = 0.286 \times 30$$

$$S = 8.57$$

For input pixel $r = 70$

$$S = 20$$

For input pixel $r = 90$

$$S = 3(90) - 190$$

$$= 270 - 90$$

$$S = 80$$

For input pixel $r = 110$

$$S = 3(110) - 190$$
$$= 330 - 190$$
$$\boxed{S = 140}$$

For input pixel $r = 200$

$$S = 230 + 0.2174\,(200 - 140)$$
$$= 230 + 0.2174\,(60)$$
$$= 230 + 13.044$$
$$\boxed{S = 243.04}$$

For input pixel $r = 140$

$$\boxed{S = 230}$$

(c) Calculate contrast gain factor

original band width:

$$140 - 70 = 70$$

Transformed band width:

$$230 - 20 = 210$$

Contrast gain:- $G = \frac{210}{70}$

$$\boxed{G = 3}$$

(d) Identify Compression and expansion.

Segment	slope	Effect
0-70	0.286	Compression
70-140	3	Expansion
140-255	0.2174	Compression.

(e) Assess visibility of diagnostic region

The transformation improves visibility because the diagnostic region 70-40 is expanded into 80-230 The contrast gain factor is 3, which makes small differences in x-ray intensity easier to identify.

3,A) Spatial Filtering - Impulse Noise removal:-

Given 3x3 Neighbourhood:

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

Centre pixel = 255 (noise impulse)

(a) calculate 3x3 mean (box) filter

$$\text{Mean} = \frac{\text{sum of all pixels}}{9}$$

$$= \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$= \frac{699}{9}$$

$$\boxed{\text{mean} = 77.67}$$

(b) Calculate weighted - average filter

Given mask:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{16}\left[(52+58+53+59) + 2(55+54+57+56) + 4(255)\right]$$

$$= \frac{1}{16}\left[222+444+1020\right]$$

$$= \frac{1686}{16}$$

$$= \boxed{105.375}$$

(c) Calculate median filter

Arrange all pixels in ascending order:

52, 53, 54, 55, 56, 57, 58, 59, 255

There are 9 values.

So, median = middle value (5th value)

$$\boxed{\text{median} = 56}$$

(d) Calculate absolute error

First, Calculate the noise-free reference using eight neighbours.

$$\text{Reference} = \frac{52+55+58+54+57+53+56+59}{8}$$

$$= \frac{444}{8}$$

$$\boxed{\text{Reference} = 55.5}$$

Absolute error formula:

$$\boxed{\text{Error} = |\text{output} - \text{Reference}|}$$

Mean filter:-

$= |77.67 = 55.5|$

Mean filter	= 22.17

weighted-average filter:- $|105.375 - 55.5| = 49.875$

median filter:- $|56 - 55.5| = 0.5$

(e) which filter is most appropriate?

The median filter is the most appropriate filter for removing salt-and-pepper noise.

Reasons:

1. It removes extreme noise values.
2. It replaces the noisy pixel with the median value.

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4)A) Edge Detection- Sobel and Prewitt Gradient Analysis:-

Given:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

(a) Calculate G_x and G_y using Sobel operator.

Sobel masks:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Calculate G_x :-

$$G_x = (-1)(10) + (0)(10) + (1)(10) + (-2)(10) + 0(10) + 0(10)$$
$$+(2)(90) + (-1)(10) + (0)(90) + (1)(90)$$
$$= -10 + 10 - 20 + 180 - 10 + 90$$

$$\boxed{G_x = 240}$$

Calculate G_y :-

$$G_y = (-1)(10) + (-2)(10) + (-1)(10) + (2)(90) + (1)(90)$$
$$= -10 - 20 - 10 + 10 + 180 + 90$$

$$\boxed{G_y = 240}$$

(b) Calculate gradient magnitude.

$$M = \sqrt{G_x^2 + G_y^2}$$
$$= \sqrt{(240)^2 + (240)^2}$$
$$= \sqrt{115200}$$

$$\boxed{M = 339.41}$$

Approximation:-

$$M \approx |G_x| + |G_y|$$
$$= 240 + 240$$

$$\boxed{M = 480}$$

Calculate percentage error :-

$$\text{Error} = \frac{|\text{Approximate} - \text{Exact}|}{\text{Exact}} \times 100$$

$$= \frac{|480 - 339.41|}{339.41} \times 100$$

$$\boxed{\text{Error} = 41.42\%}$$

(c) Calculate gradient direction.

$$\boxed{\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right)}$$

$$= \tan^{-1}\left(\frac{240}{240}\right)$$

$$\boxed{\alpha = 45^\circ}$$

The gradient direction is 45°. The edge orientation is perpendicular to the gradient. i.e. 135° (equivalently -45°).

(d) Repeat using Prewitt operator.

Prewitt masks:-

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Calculate G_x:-

$$G_x = -10+10-10+90-10+10-10+90-10+90$$

$$\boxed{G_x = 160}$$

Calculate G_y:

$$G_y = -10-10-10+10+90+90$$

$$\boxed{G_y = 160}$$

Gradient magnitude:-

$$M = \sqrt{(160)^2 + (160)^2}$$

$$= \sqrt{51200}$$

$$\boxed{M = 226.27}$$

Approximation:-

$$M \approx 160 + 160$$

$$\boxed{M = 320}$$

Percentage error:- $= \frac{320 - 226.27}{226.27} \times 100$

$$= \boxed{41.42\%}$$

(e) Determine whether each operator detects an edge.

Given threshold:

$M \geq 250$

operator	Gradient magnitude	Threshold	Result
Sobel	339.41	250	Edge detected
Prewitt	226.27	250	Edge not detected.

Explanation:-

The sobel operator detects the edge because its gradient magnitude is greater than 250.

The prewit operator doesnot detect the edge because its gradient magnitude is less than 250.

5)A) Global thresholding - Iterative and otsu segmentation

Given:

3-bit image, $L=8$

Total pixels = 32

Intensity (r_k) : 0, 1, 2, 3, 4, 5, 6, 7

Frequency(n_k) : 3, 6, 4, 1, 1, 5, 8, 4

(a) calculate global mean intensity

$$m_G = \frac{\sum r_k n_k}{N}$$

Substitute :- $\Rightarrow \frac{(0)(3)+(1)(6)+(2)(4)+(3)(1)}{32} + \frac{(4)(1)+(5)(5)+(6)(8)+(7)(4)}{32}$

$$= \frac{122}{32}$$

$$\boxed{m_G = 3.8125}$$

(b) Basic iterative global thresholding

Initial threshold:

$$T_0 = m_G = 3.8125$$

Divide pixels into two groups:

→ Group 1: Intensities less than (or) equal to T.

→ Group 2: Intensities greater than T.

Iteration 1:

Group 1 : 0, 1, 2, 3

Group 2 : 4, 5, 6, 7

Calculate first group mean:

$$m_1 = \frac{(0)(3)+(1)(6)+(2)(4)+(3)(1)}{3+6+4+1}$$

$$= \frac{17}{14} = 1.2143$$

Compute second group mean:

$$m_2 = \frac{(4)(1)+(5)(5)+(6)(8)+(7)(4)}{1+5+8+4}$$

$$= \frac{105}{18} = 5.8333$$

Updated threshold:-

$$T_1 = \frac{m_1+m_2}{2} = \frac{1.2143+5.8333}{2}$$

$$T_1 = 3.5238$$

Calculate threshold difference :-

$$\Delta T = |T_1 - T_0|$$
$$= |3.5238 - 3.8125|$$

$$\boxed{\Delta T = 0.2887}$$

Since :

$$0.2887 < 0.5$$

The stopping condition is satisfied.

(c) Calculate between-class variance using otsu's method.

$$\sigma_B^2(k) = P_1(k)\, P_2(k) [m_1(k) - m_2(k)]^2$$

where :

→ P_1 = Probability of first class.

→ P_2 = probability of second class.

→ m_1 = mean of first class.

→ m_2 = mean of second class.

For k=2 :

class $c_1 = 0, 1, 2$

class $c_2 = 3, 4, 5, 6, 7$

$$P_1 = \frac{13}{32} = 0.40625$$

$$P_2 = \frac{19}{32} = 0.59375$$

$$m_1 = \frac{14}{13} = 1.0769$$

$$m_2 = \frac{108}{19} = 5.6842$$

$$\sigma_B^2(2) = 0.40625(0.59375)(1.0769 - 5.6842)^2$$

$$\boxed{\sigma_B^2(2) = 5.1202}$$

For $k=3$:

Class $c_1 = 0,1,2,3$

Class $c_2 = 4,5,6,7$

$$P_1 = \frac{14}{32} = 0.4375$$

$$P_2 = \frac{18}{32} = 0.5625$$

$$m_1 = \frac{17}{14} = 1.2143$$

$$m_2 = \frac{105}{18} = 5.8333$$

$$\sigma_B^2(3) = 0.4375(0.5625)(1.2143 - 5.8333)^2$$

$$\sigma_B^2(3) = 5.2506$$

For $k=4$:

Class $c_1 = 0,1,2,3,4$

Class $c_2 = 5,6,7$

$$P_1 = \frac{15}{32} = 0.46875$$

$$P_2 = \frac{17}{32} = 0.53125$$

$$m_1 = \frac{21}{15} = 1.4$$

$$m_2 = \frac{101}{17} = 5.9412$$

$$\sigma_B^2(4) = 0.46875(0.53125)(1.4 - 5.9412)^2$$

$$\boxed{\sigma_B^2(4) = 5.1354}$$

(d) Determine optimum otsu threshold.

The between-class variance is maximum at $k=3$

Therefore,

$$\boxed{k^* = 3}$$

Binary mapping:

→ Intensities 0, 1, 2, 3 → object (1)

→ Intensities 4, 5, 6, 7 → Background (0)

Percentage of object pixels : $\frac{14}{32} \times 100 = \boxed{43.75\%}$

(e) Compare both thresholding methods.

→ Iterative threshold = 3.5238

→ otsu threshold = 3.

→ Both methods give similar threshold values.

→ The histogram is suitable for global thresholding because it seperates ~~blac~~ dark objects from the bright background.